

Batch: A2 Roll No.: 16010324044

Experiment No. 3

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

TITLE: Linear convolution, Circular convolution, Auto correlation and cross correlation of sequences. Applications of correlation.

AIM: 1) To find linear convolution, Circular convolution, Auto correlation and cross correlation of given sequences/signals.
2) Applications of correlation

OUTCOME: Analyse discrete time systems in time and Z-domain.

PROCEDURE:

1. Read Discrete time signal $x(n)$
2. Read Impulse response signal $h(n)$
3. Perform linear convolution to get output $y(n)$
4. Perform linear convolution to get output $y(n)$
5. Read Discrete time signal $x_1(n)$
6. Read Discrete time signal $x_2(n)$
7. Perform Autocorrelation and cross correlation on signals
8. Plot or note responses

Analytical solutions:

```
clc;
close all;
clear all;
%% Linear and Circular Convolution
x = input('Enter discrete time input sequence x = ');
h = input('Enter impulse response sequence h = ');
yl = conv(x,h);
yc = cconv(x,h,max(length(x),length(h)));
xl = 0:length(x)-1;
hl = 0:length(h)-1;
yll = 0:length(yl)-1;
ycl = 0:length(yc)-1;
figure;
subplot(5,1,1)
stem(xl,x,'filled')
title('Input Sequence x(n)')
xlabel('n')
ylabel('Amplitude')
subplot(5,1,2)
stem(hl,h,'filled')
title('Impulse Response h(n)')
xlabel('n')
ylabel('Amplitude')
subplot(5,1,3)
stem(yll,yl,'filled')
title('Output by Linear Convolution')
xlabel('n')
```

```

ylabel('Amplitude')
subplot(5,1,4)
stem(ycl,yc,'filled')
title('Output by Circular Convolution')
xlabel('n')
ylabel('Amplitude')
% Zero padding and Linear Convolution
zx = [x zeros(1,length(yl)-length(x))];
zh = [h zeros(1,length(yl)-length(h))];
ycm = cconv(zx,zh,length(yl));
subplot(5,1,5)
stem(yll,ycm,'filled')
title('Circular Convolution using Zero Padding')
xlabel('n')
ylabel('Amplitude')
%% Auto Correlation and Cross Correlation
x1 = input('Enter first sequence x1 = ');
x2 = input('Enter second sequence x2 = ');
at1 = xcorr(x1);
at2 = xcorr(x2);
ct = xcorr(x1,x2);
l1 = -(length(x1)-1):(length(x1)-1);
l2 = -(length(x2)-1):(length(x2)-1);
lc = -(max(length(x1),length(x2))-1):(max(length(x1),length(x2))-1);
figure;
subplot(3,1,1)
stem(l1,at1,'filled')
title('Auto Correlation of x1')
xlabel('Lag')
ylabel('Amplitude')
subplot(3,1,2)
stem(l2,at2,'filled')
title('Auto Correlation of x2')
xlabel('Lag')
ylabel('Amplitude')
subplot(3,1,3)
stem(-(length(x2)-1):(length(x1)-1),ct,'filled')
title('Cross Correlation of x1 and x2')
xlabel('Lag')
ylabel('Amplitude')

```

Input:

Command Window

```

Enter discrete time input sequence x = [1, 2, 3, 4]
Enter impulse response sequence h = [2, 3, 4, 2]
Enter first sequence x1 = [1, 2, 3, 4]
Enter second sequence x2 = [2, 3, 4, 2]
>>

```

Output:

Figure 1 ×

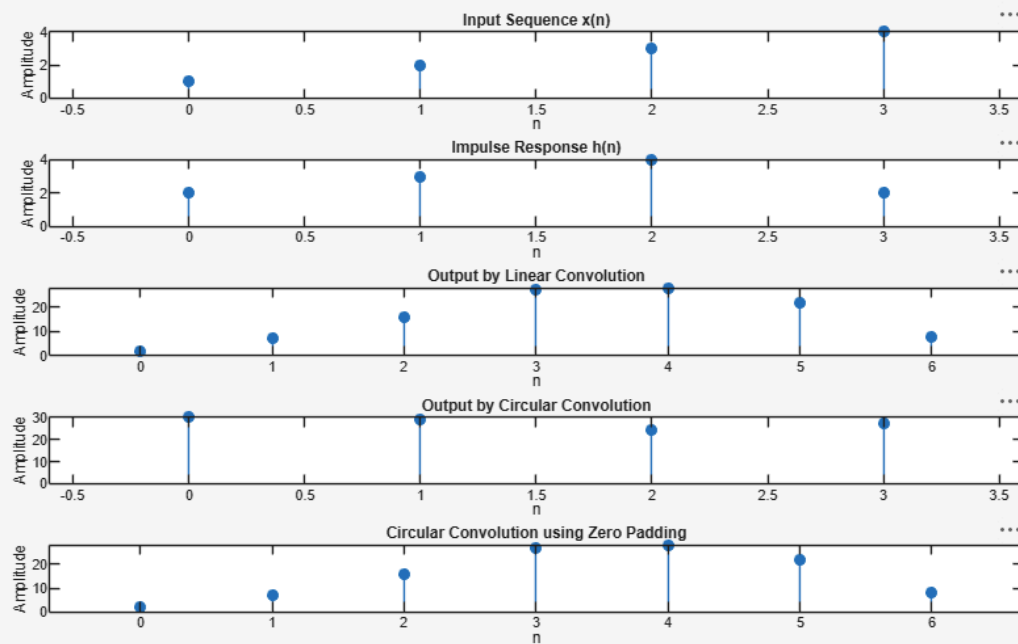
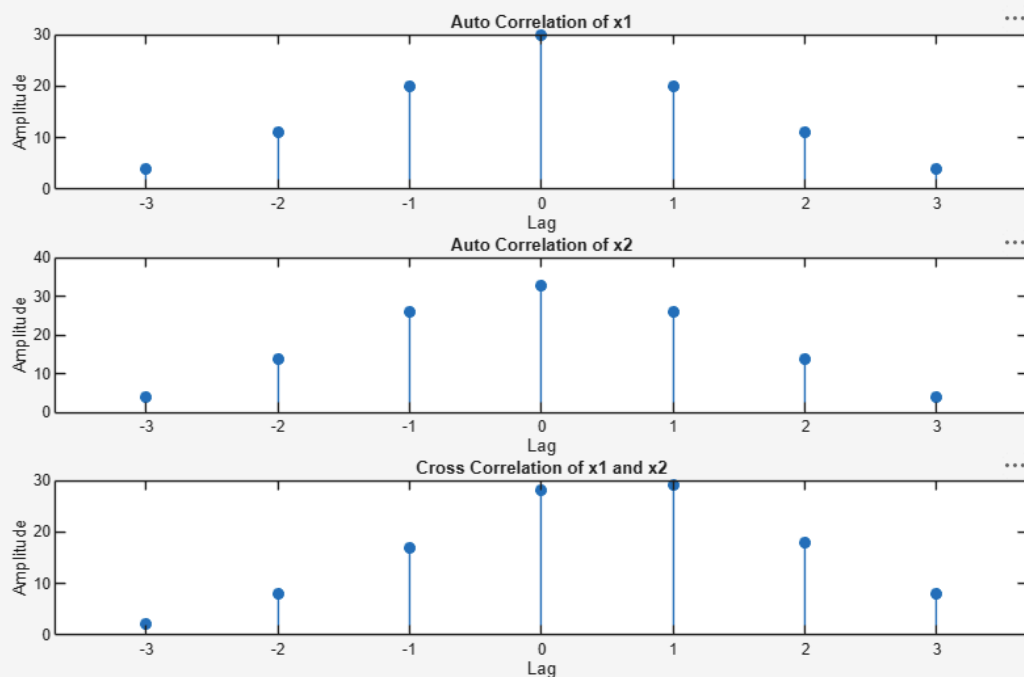


Figure 1 × Figure 2 ×



Applications of Correlation:

- **Signal Detection:** Correlation is used to detect a known signal in the presence of noise.
- **Pattern Recognition:** It helps in identifying and matching similar patterns or signals.
- **Time Delay Estimation:** Cross-correlation can be used to determine the time delay between two signals.
- **Radar and Sonar Systems:** Correlation is used to detect and locate targets by comparing transmitted and received signals.
- **Communication Systems:** It is used for synchronization, signal recovery and detection of transmitted data.
- **Speech and Audio Processing:** Correlation helps in speech analysis, pitch detection and identifying similarities between audio signals.
- **Image Processing:** Correlation is used for template matching and object detection.

CONCLUSION

In this experiment, I successfully performed the experiment to study linear convolution, circular convolution, autocorrelation and cross-correlation of discrete-time sequences using MATLAB. The output responses obtained from the plots verified the convolution and correlation operations. Autocorrelation was used to study the similarity of a signal with its delayed version, while cross-correlation was used to determine the similarity between two different signals at different lags. Thus, the experiment provided a practical understanding of convolution and correlation and their applications in digital signal processing.

Signature of faculty in-charge